

	Nutan Maharashtra Vidya Prasarak Mandal's (NMVPM's)			 ESTD : 1906
	NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING AND TECHNOLOGY			
	Under Administrative Support - Pimpri Chinchwad Education Trust (PCET)			
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"Samarth Vidya Sankul", Vishnupuri, Telegaon Dabhade, Taluka Maval, District Pune - 410507				
Tel. No. 02114 - 231666		E-mail : nmietdalegaon@gmail.com		Web : www.nmiet.edu.in

Computer Engineering

ASSIGNMENT

Academic Year: 2025-26 (Semester II)

YEAR/ SEM-TE SEM-II

MAX.MARKS- 10 Per Unit (10*6)

SUBJECT: Data Science and Big Data Analytics

CO	BloomsLevel (1-6)	Q.N.	Questions
Assignment 1			
1	L1	1	What is Data wrangling? Explain methods in detail.
1	L2	2	Describe 5 V's of Big Data in detail.
1	L2	3	Explain Data Science Life Cycle.
1	L4	4	Case Study Question: A leading e-commerce company is experiencing rapid growth in user activity, resulting in massive data generation from customer transactions, product reviews, clickstreams, and social media interactions. The company wants to leverage this data to improve customer experience, optimize pricing strategies, and enhance recommendation systems. However, the collected data is highly unstructured, incomplete, and inconsistent. The organization currently uses traditional Business Intelligence (BI) tools but is facing challenges in handling large-scale, real-time data and extracting meaningful insights. a) Explain why the company needs Data Science and Big Data technologies in this scenario. b) Identify and explain the 5 V's of Big Data with reference to the case. c) Describe the Data Science Life Cycle steps that the company should follow to extract insights from its data.

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Assignment 2

2	L2	1	Explain Measures of Central Tendency: Mean, Median, Mode, Mid-range with example.
2	L2	2	Explain hypothesis and hypothesis testing with example.
2	L2	3	Elaborate Chi-Square Tests with example.
2	L4	4	Case Study: Online Retail Data Analysis An e-commerce company collects daily transaction data from its platform. The dataset includes customer purchase amounts, product categories, and customer demographics. The company wants to analyze customer behavior and improve decision-making using statistical methods. A sample dataset of purchase amounts (in ₹) for a day is as follows: [250, 300, 450, 500, 300, 700, 800, 300, 450, 600] Additionally: The company wants to understand variability in purchases. It aims to test whether a new recommendation system has significantly increased average purchase value. It also wants to check whether customer category (New/Returning) is independent of product category purchased. a) Calculate Mean, Median, and Mode of the purchase amounts. b) Interpret which measure best represents customer spending behavior and why. c) Calculate Range and Standard Deviation for the dataset. d) Explain how Pearson Correlation can be used to analyze the relationship between time spent on the website and purchase amount

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Assignment 3

3	L2	1	Describe all six phases of the Data Analytic Lifecycle in detail.
3	L2	2	Define Big Data and explain its importance in modern applications.
3	L2	3	List and explain sources of Big Data with real-life examples.
3	L2	4	Explain how Big Data analytics can be applied in: • Smart Cities • Healthcare • Banking
3	L4	5	Case Title: A healthcare organization collects patient data from wearable devices, hospital records, and mobile apps. Explain how the Data Analytic Lifecycle can be applied to predict diseases and improve patient care.

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Assignment 4

4	L2	1	Discuss different techniques for handling missing data.
4	L2	2	Explain the difference between Linear Regression and Logistic Regression with examples.
4	L2	3	Explain Decision Tree algorithm with a suitable example.
4	L2	4	What are the different types of analytics? Explain with real-life use cases.
4	L4	5	Case Study: Online Retail Data Analysis System An e-commerce company wants to analyze its customer transaction dataset to improve sales and recommendations. The dataset contains the following issues: • Duplicate transaction records • Missing values in customer age and product category • Inconsistent price formats • Customer purchase history The company wants to: • Clean and preprocess the data • Analyze buying patterns • Predict whether a customer will purchase a product • Recommend frequently bought items together a) Describe how to handle missing values in customer age and product category. b) Explain how the Apriori Algorithm can be used to find frequent item sets. c) Which regression technique will you use to predict customer spending amount? Why?

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Assignment 5			
5	L1	1	Define Clustering and differentiate between K-Means and Hierarchical Clustering
5	L1	2	Define: - Precision - Recall - F1-score
5	L2	3	Explain Holdout Method and Random Subsampling with examples
5	L3	4	Draw and explain Confusion Matrix for binary classification
5	L4	5	Case Study: A retail company wants to improve its marketing strategy using customer data. The dataset contains: - Customer purchase history (time-series data) - Product reviews (text data) - Customer segmentation data (age, income, spending score) The company aims to: - Segment customers into groups - Analyze product reviews - Predict customer behavior - Evaluate model performance - Explain how K-Means clustering can be applied to segment customers. - What is the role of the Elbow Method in selecting K? - For predicting customer purchase behavior:- Explain Confusion Matrix - The dataset includes time-based purchase history.- Explain how Time-Series Analysis can help in forecasting

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Assignment 6			
6	L1	1	Define Data Visualization and explain its importance in Big Data.
6	L2	2	Explain various data visualization techniques used in analytics.
6	L2	3	Explain the architecture of Apache Hadoop .
6	L4	4	Compare Hadoop, Hive, and Pig.
6	L4	5	Case Study A large e-commerce company, Amazon, generates massive volumes of data daily from customer transactions, product searches, reviews, and logistics operations. The company wants to: - Analyze customer buying patterns - Identify peak sales periods - Detect anomalies in transactions - Improve recommendation systems Due to the huge volume, variety, and velocity of data, traditional visualization techniques are not sufficient. The company adopts Big Data technologies like Apache Hadoop, Apache Hive, and Apache Pig to process and visualize the data efficiently. They also use Python-based visualization tools such as Matplotlib and Seaborn for graphical representation. a) Explain the challenges faced in Big Data Visualization in the above case study. Suggest suitable solutions. b) Discuss the analytical techniques used in Big Data visualization for business decision-making.